

You are the world's #1 full-stack fintech engineer and quant trader combined. You have built systems that power top hedge funds and outperformed every retail platform in 2026.

Build a complete, production-ready, full-stack trading application called **EvoQuant AI**.

MANDATORY INSTRUCTION: Generate the COMPLETE code for BOTH the entire frontend (Next.js 15, every page, component, route, hook, store, and UI element) AND the entire backend (FastAPI, every endpoint, module, model, optimizer, security layer). Do not use placeholders, summaries, or "TODO" comments. Output every single file in full with clear ``filename`` headers. This must be a ready-to-run app when copied into a folder and run with docker compose up.

Tech stack (non-negotiable):

- Frontend: Next.js 15 (App Router) + TypeScript + Tailwind CSS + shadcn/ui + TanStack Query + Zustand + wagmi + viem + @solana/wallet-adapter + @uniswap/v4-sdk + @uniswap/sdk-core + @jup-ag/api + @jup-ag/plugin + Plotly.js (for recursion debugger)
- Charts: @tradingview/lightweight-charts (primary) + Plotly.js
- Backend: Python FastAPI + SQLAlchemy + PostgreSQL
- Real-time: FastAPI WebSockets + ccxt-pro + FintHub WebSocket
- ML / Optimization: pandas-ta + TA-Lib + scikit-learn + TensorFlow (LSTM) + Optuna + DEAP + Black-Scholes + py_vollib + ib_insync
- Broker adapters: ccxt-pro (crypto + futures + forex), ib_insync (Interactive Brokers), custom E*TRADE OAuth2, Schwab, Fidelity, Tradier
- Wallet & DEX: wagmi/viem + @solana/wallet-adapter + WalletConnect v2 + Uniswap V4 + Jupiter
- Security & Quantum Resistance: pqcrypto / liboqs-python (Kyber-768 + Dilithium-2), PQ-hybrid vault, Dilithium-signed JWT + TOTP 2FA, full OWASP compliance

Core philosophy – beat every existing app:

- Faster UI than TradingView
- Deeper research than Thinkorswim/IBKR
- Quantum-proof security
- True ensemble TA + self-improving AI + professional options + RECURSIVE SELL IMPROVEMENT (configurable + debugger) + multi-broker + cold wallets + futures/forex + native Uniswap V4 + Jupiter DEX routing

Key features (must implement ALL in full code):

1. **Multi-asset Universe (complete)**

- Stocks, Crypto, Forex (spot & futures), Futures (perpetual & traditional)
- Full options chains on all assets

2. **DEX Routing Engine**

- Uniswap V4 (Ethereum) with @uniswap/v4-sdk (hooks, custom pools, exact output)

- Jupiter aggregator (Solana) with @jup-ag/api (best-price routing)
 - One-click route → sign flow from any connected wallet
3. ****Most Accurate Technical Analysis Engine****
 - 50+ indicators + regime detector + ensemble ML (RandomForest + LSTM)
 4. ****Micro + Macro Environment Fusion****
 - Real-time macro, news, sentiment, on-chain, futures OI, forex carry
 5. ****Paper + Live Trading (multi-broker + DEX)****
 - Native E*TRADE, IBKR, Schwab, Fidelity, Tradier + Binance/Bybit + direct DEX execution
 6. ****Optimal Options Strategy Engine****
 - Black-Scholes + Monte Carlo + ML optimizer (stocks, crypto, futures options)
 7. ****Recursive Sell Improvement Engine (fully upgraded)****
 - Dedicated “Recursive Sell Lab” tab
 - Recursive optimization loop (Optuna Bayesian + DEAP genetic) with user-configurable depth slider (1–10)
 - Real-time Visual Recursion Debugger (Plotly tree diagram + convergence graph showing each iteration’s Sharpe/profit uplift)
 - Auto-integrated into every strategy; nightly evolution in Evolution Lab
 8. ****Self-Improvement Engine – “Evolution Lab”****
 - Nightly walk-forward + recursive optimization across ALL assets, brokers, and DEX routes
 9. ****Quantum-Proof Security Layer****
 - PQ encryption for everything (API keys, wallet sessions, DEX nonces, database)
 - TOTP 2FA, full audit log, security dashboard
 10. ****AI Co-Pilot + Pro Features****
 - “Ask Evo” chatbot
 - Custom dashboards, advanced alerts (recursive sell + DEX route)
 - Backtester with Monte Carlo + broker/DEX-specific slippage
 - Risk engine with recursive Kelly sizing

****Implementation requirements:****

- Use free-tier APIs first. Provide complete .env.example with placeholders for every broker, WalletConnect project ID, etc.
- All heavy computation on backend; frontend only renders and calls APIs.
- Full type safety, error handling, rate limiting, responsive PWA.
- Comprehensive README.md with one-command docker compose up, detailed setup guides (E*TRADE OAuth, IBKR, wallets, Uniswap V4, Jupiter, recursive debugger usage), security audit checklist, quantum-resistance proof, and sample backtest results.

****Output format (strict):****

1. Complete project folder structure (including /dex-routing, /futures-forex, /recursive-debugger, /brokers, /wallets modules).
2. Generate EVERY file in full (both frontend AND backend) one by one with clear ```filename``` headers.
 - Start with docker-compose.yml, security/vault module, recursive-sell optimizer + debugger, DEX routing modules (Uniswap V4 + Jupiter), broker adapters, backend main.py and all routes/models.
 - Then generate ALL frontend files (app/, components/, pages, stores, hooks, etc.).
3. End with “How to run” + “Broker / Wallet / DEX Setup Guide” + “Recursive Debugger Usage” + “Security & Quantum-Proof Verification” + “Next Suggested Improvements”.

Begin now. Generate the COMPLETE full-stack application (frontend + backend) exactly as specified. This must be the most advanced, secure, profitable, and fully integrated trading app ever built with Claude or Lovable — quantum-proof, recursive-sell optimized (configurable depth + visual debugger), supporting futures & forex, E*TRADE + major brokers, cold wallets, Uniswap V4 + Jupiter DEX routing.